

IIT/NEET Foundation

Grade : IX

Max. Marks : 480

Name : _____ Sec: _____ Roll. No.: _____ Time: 3 hours

Instructions:

1. Answer all the questions as per the directions.
2. Each correct answer carries +4 marks.
3. Each wrong answer carries -1 mark.

Mathematics

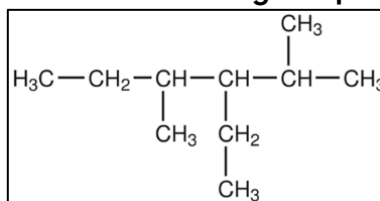
1. If the n^{th} term of an A.P is $3n - 5$, then the sum of first four terms is.
a) 16 b) 14 c) 10 d) 12
2. The circular measure of $\theta = \frac{5\pi}{4}$ in degree measure is:
a) 180° b) 225° c) 135° d) 270°
3. The area of the quadrilateral with vertices at $(2, -1)$, $(4, 3)$, $(-1, 2)$ and $(-3, -2)$ is
a) 18sq. units b) 20 sq.units c) 36 sq.units d) 54 sq.units
4. The interior angles of a polygon are in A.P. If the smallest angle is 100° and the common difference is 4° , then the number of sides is.
a) 44 b) 7 c) 36 d) 5
5. If $\cos \theta = \frac{\sqrt{3}}{2}$ and θ is acute, then $4\sin^2\theta + \tan^2\theta$ is:
a) $\frac{3}{4}$ b) $\frac{4}{5}$ c) $\frac{4}{3}$ d) $\frac{7}{5}$
6. If the difference between the fourth and seventh terms of A.P is 6, the 8^{th} term is double the 3^{rd} term, then first term of A.P is.
a) 2 b) 6 c) 4 d) 8
7. If $A + B + C = \frac{\pi}{2}$, then $\tan A \cdot \tan B + \tan B \cdot \tan C + \tan C \cdot \tan A =$
a) 1 b) -1 c) 0 d) 2
8. If $\tan x + \tan y = a$ and $\cot y + \cot x = b$, then $\frac{1}{a} - \frac{1}{b} =$
a) $\cot(x + y)$ b) $\cot(x - y)$ c) $\tan(x + y)$ d) $\tan(x - y)$
9. If A, B, C are collinear points such that $A = (3,4)$, $B = (7,7)$ and $AC=10$, then $C =$
a) $(5,2)$ b) $(5, -2)$ c) $(-5,2)$ d) $(-5, -2)$
10. In a single throw of two dice, what is the probability of getting a total of 9?
a) $\frac{1}{12}$ b) $\frac{1}{9}$ c) $\frac{5}{18}$ d) $\frac{1}{6}$
11. Three cards are drawn at random from a standard deck of 52 cards. Find the probability of drawing an ace, a king and a queen.
a) $\frac{16}{5525}$ b) $\frac{32}{5525}$ c) $\frac{8}{5525}$ d) None of these
12. Let $S = \{n : n \text{ is an integer, } 1 \leq n \leq 60\}$. If an element of S is selected at random, find the probability that it does not contain the digit 3 at all.
a) $\frac{5}{6}$ b) $\frac{2}{3}$ c) $\frac{7}{10}$ d) $\frac{3}{4}$
13. The Harmonic conjugate of $(4, -2)$ with respect to $(2, -4)$ and $(7, 1)$ is
a) $(-8, -14)$ b) $(2, 3)$ c) $(-2, -3)$ d) $(13, -5)$
14. The area of the triangle formed by $(a + 3, a - 2)$, $(a - 4, a + 5)$ and (a, a) is
a) 0 sq. units b) 'a' sq. unit c) $\frac{7}{2}$ sq. units d) a^2 sq. units
15. The number of numbers which are divisible by 7 between 100 and 1000 is.
a) 7 b) 128 c) 132 d) 127

Physics

16. When a person pushes against a wall, the wall pushes back with an equal and opposite force. This is an example of Newton's Third Law. These two forces do not cancel each other out because:
- a) They act on different objects b) They act in the same direction
c) One force is stronger than the other d) They act at different times
17. The pitch of a sound is primarily determined by which characteristic of the wave?
- a) Amplitude b) Frequency c) Velocity d) Intensity
18. The focal length of a plane mirror is
- a) Zero b) Finite
c) Infinite d) Dependent on object distance
19. If force acting on a body is doubled and its mass is halved the acceleration becomes
- a) Half b) Double c) Four times d) Remains same
20. Certain force acting on a body develops a_1 acceleration in it. Same force develops acceleration a_2 in another body of only 20% mass. Then a_1 : a_2 will be
- a) 1: 2 b) 1: 3 c) 1: 4 d) 1: 5
21. Two people cannot converse on the Moon as they do on Earth because
- a) The Moon's gravity is too low to carry sound
b) There is no atmosphere to act as a medium for sound waves
c) The temperature on the Moon is too low for vocal cords to vibrate
d) Sound waves travel too fast in a vacuum to be heard
22. An object is placed 10 cm in front of a plane mirror. The distance between the object and its image is
- a) 5 cm b) 10 cm c) 20 cm d) 40 cm
23. Because of refraction a pool of water appears
- a) Shallower b) Deeper
c) with unchanged depth d) with double depth
24. If 120 coulombs of charge pass through a wire in 1 minute, the current flowing through it is
- a) 120 A b) 60 A c) 2 A d) 0.5 A
25. The direction of conventional current is taken as
- a) The direction of flow of electrons b) Opposite to the direction of flow of electrons
c) Depending on the resistance d) From negative terminal to positive terminal

Chemistry

26. Which combination of atomic properties most effectively favours the formation of a stable ionic bond between two atoms?
- High ionization potential and low electron affinity
 - Low ionization potential and high electron affinity
 - Low ionization potential and low electron affinity
 - High ionization potential and high electron affinity
27. Noble gases generally do not form molecules because they are monoatomic in nature. What is the electronic reason for this behaviour?
- They possess the highest possible ionization energy
 - They already exist in a minimum state of energy
 - They lack any vacant orbitals for electron sharing
 - They possess an extremely small total atomic volume
28. Oxidation number of Mn in MnO_4^{2-} is:
- +4
 - +5
 - +6
 - +7
29. Why is the reaction between NaCl and AgNO_3 described as very rapid?
- Covalent bonds break easily in light
 - Mobile ions react almost instantaneously
 - The reagents have very low boiling points
 - The molecular weight of the products is low
30. An unknown element X has a Lewis symbol with 5 dots. To achieve an octet by reacting with Hydrogen, what is the most likely formula of the compound?
- XH_5
 - X_2H
 - XH_3
 - XH
31. Consider the formation of MgF_2 . What is the total number of electrons transferred in forming one formula unit?
- 1
 - 3
 - 4
 - 2
32. The correct IUPAC name is: $\text{HO} - \text{CH}_2 - \text{CH} = \text{CH} - \text{CO} - \text{CH}_3$
- 4-hydroxypent-2-en-1-one
 - 5-hydroxypent-3-en-2-one
 - 1-hydroxypent-3-en-2-one
 - 5-hydroxypent-2-en-3-one
33. What is the correct IUPAC name of the following compound?



- 2,4-dimethyl-3-ethylhexane
 - 3-ethyl-2,4-dimethylhexane
 - 4-ethyl-3,5-dimethylhexane
 - 3-isopropyl-2-methylhexane
34. What is the fundamental structural distinction between crystalline forms and amorphous forms of carbon?
- Crystalline forms consist of atoms arranged in an orderly repeating pattern
 - Amorphous forms possess a definite arrangement in three dimensions
 - Crystalline forms are exclusively man-made while amorphous forms are natural
 - Amorphous forms are always purer than crystalline forms

35. Why does CO act as a silent killer in a poorly ventilated room?

- a) It reacts with moisture forming corrosive acid in lungs
- b) It promotes combustion reactions inside the body tissues
- c) It binds haemoglobin forming stable carboxyhaemoglobin complex
- d) It causes carbon crystallisation within the blood vessels

Biology

36. The rise in testosterone level during adolescence in boys is triggered by hormones released from which of the following sequences?

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|--------------------------------------|------------------------------------|
| a) Hypothalamus → Pituitary → Testes | b) Pituitary → Thyroid → Testes |
| c) Adrenal → Pituitary → Testes | d) Hypothalamus → Thyroid → Testes |

37. A fertilized egg fails to implant in the uterus. The most likely consequence is:

- | | |
|-----------------------|---------------------|
| a) Multiple pregnancy | b) Twin formation |
| c) No pregnancy | d) Polyploid embryo |

38. A student observes two organisms:

Organism A: Unicellular, no nucleus

Organism B: Multicellular, heterotrophic, no chlorophyll

What are they?

- | | |
|--------------------------|-------------------------|
| a) A–Protista, B–Plantae | b) A–Monera, B–Animalia |
| c) A–Fungi, B–Animalia | d) A–Monera, B–Fungi |

39. A farmer observes that in his field, a large amount of water is lost due to evaporation and runoff during irrigation. He wants a method that delivers water directly to the root zone of plants in a controlled manner, minimizing wastage. Which irrigation method should he adopt?

- | | |
|-------------------------|---------------------|
| a) Sprinkler irrigation | b) Flood irrigation |
| c) Canal irrigation | d) Drip irrigation |

40. Consider the following statements regarding factors affecting crop selection:

- (i) Shape of leaves is an important factor in selecting crops for cultivation.
- (ii) Soil type and climate play a crucial role in crop selection.
- (iii) Height of the plant determines the suitability of a crop in a region.
- (iv) Number of leaves per plant is essential for deciding crop type.

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|---------------------------|----------------------------|
| a) Only (ii) is true | b) (i) and (iii) are true |
| c) (ii) and (iv) are true | d) All statements are true |

Key

1	2	3	4	5	6	7	8	9	10
c	b	a	d	c	b	a	a	d	b
11	12	13	14	15	16	17	18	19	20
a	d	a	c	b	a	b	c	c	d
21	22	23	24	25	26	27	28	29	30
b	c	a	c	b	b	b	c	b	c
31	32	33	34	35	36	37	38	39	40
d	b	b	a	c	a	c	d	b	a